

Benrabah Seif Islem

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About Me

Full-stack developer, AI enthusiast & Designer passionate about building intelligent and scalable solutions. With experience in Machine Learning, Deep Learning, and modern web development, I create applications that combine technology and design to deliver intelligent user experiences.

Education

Higher School of Computer Science 08 May 1945 – Sidi Bel Abbes (ESI SBA) Sidi Bel Abbes, Algeria
Software engineering Master degree | Engineer Diploma in software engineering 2023 – Currently

Mohamed Boudiaf University

M'sila, Algeria

License 2 Computer Science

2021 – 2023

- **Awards/Honors:** Graduated among the top students, class of 2021

Experience

IT Internship: Algérie Telecom – M'sila

Sept – October 2024

- Assisted in diagnosing and resolving system and network issues while supporting internal users and monitoring connectivity across departments. Documented troubleshooting procedures and contributed to improving workflow efficiency.

Projects

Resynex | *AI-Powered Clinical Research Platform* / *Full-Stack & AI/ML* github.com/SeifIslemBenrabah/Resynex-DTG

- Architected a **14-service microservices platform** for clinical research management using Spring Cloud Gateway, Netflix Eureka, React/TypeScript, Node.js/Express, and Python/FastAPI, orchestrated with Docker Compose on a Linux VPS.
- Built a **Patient Digital Twin Generator (DTG)** using real PPMI (Parkinson's disease) cohort data with **Monte Carlo uncertainty quantification**, producing 12-month disease progression forecasts with 95% confidence intervals (accuracy: 84.7%).
- Implemented **RAG knowledge notebooks** (NotesLab) using ChromaDB vector database and Google Gemini API, enabling document-grounded AI research assistance for clinical teams.
- Built **LitScope**, a unified literature search querying PubMed, Semantic Scholar, arXiv, and OpenAlex simultaneously, and a 5-analysis-type statistical module with interactive visualizations.
- Developed a real-time AI research assistant, team collaboration, and an automated demo seeding system (Quick Access) showcasing live digital twin generation.

Delivery Platform | *Full-Stack*

- Developed a smart delivery platform using Express.js, Spring Boot, and FastAPI in a microservices architecture. Integrated MongoDB and MySQL, implemented WebSockets for real-time tracking, secure authentication with JWT and OAuth, and an optimization microservice for intelligent order assignment and route optimization.

E-Learning Platform | *Full-Stack*

- Developed an online learning platform with authentication, course management, and user dashboards. Built an admin panel for teachers to manage students, grades, and modules using React, Node.js, Express, MySQL, and Sequelize.

Community Management Platform | *Full-Stack*

- Developed a community management platform with modules for user management, event scheduling, and project tracking. Implemented secure authentication, role-based access control, and dynamic dashboards using React, Node.js, Express, and MySQL.

Movies Recommendation System | *Machine Learning & Full-Stack*

- Developed a movie recommendation system using scikit-learn, pandas, and NLTK to analyze movie descriptions and suggest similar titles based on user search queries through content-based filtering.

Skills

- Machine Learning & Deep Learning
- Microservices & API Design
- Backend: Spring Boot, Express.js, FastAPI
- Frontend: React, React Native
- Authentication & Security (JWT, OAuth)
- Databases: MySQL, MongoDB, Oracle SQL
- LLMs, RAG & Vector Databases
- System Design & Scalability
- Docker & Kubernetes
- Algorithms & Data Structures
- Problem Solving & Optimization
- UI/UX Design
- Git & Linux Environments

Languages

- Arabic – Native
- French – B2
- English – B2